Batterjee Medical College
For Science and Technology

An approach to a Case of Dyspnea

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Heart Failure

Learning Objectives

Knowledge & understanding

- Define heart failure by ejection fraction (HFrEF vs. HFpEF), affected ventricle (left, right, or biventricular), and clinical course (chronic compensated vs. acute decompensated).
- Describe the ACC/AHA stages of heart failure and the Framingham diagnostic criteria for congestive heart failure.

Skill

- Apply the Framingham criteria to establish a diagnosis of congestive heart failure.
- Formulate a guideline-directed medical therapy (GDMT) regimen for a patient with HFrEF.
- Formulate an emergency management plan for a patient presenting with acute decompensated heart failure.

Main Subtopics

- Definition and classifications
- Epidemiology
- Pathophysiology and Compensatory Mechanisms
- Clinical Features
- The ACC/AHA Stages of Heart Failure
- Diagnostics :
  - Blood work, cardiac biomarkers,
  - Imaging and ECG
  - Framingham criteria
- Management:
  - Lifestyle Modifications
  - HFrEF
  - HFpEF
  - Invasive Interventions
- Subtypes and Variants (high-output heart failure)
- Acute Heart Failure as a Medical Emergency

Case scenario :

A 43-year-old man comes to the physician because of a 1-month history of increasing shortness of breath. He has been using two pillows at night but frequently wakes up feeling as if he is choking. Five months ago, he underwent surgery for the creation of an arteriovenous fistula in his left upper arm. He has hypertension and chronic kidney disease due to reflux nephropathy. He receives hemodialysis three times a week. His current medications are enalapril, vitamin D3, erythropoietin, sevelamer, and atorvastatin. His temperature is 37.1°C (98.8°F), pulse is 103/min and bounding, respirations are 22/min, and blood pressure is 106/58 mm Hg. Physical examination shows bilateral pitting pedal edema. There is jugular venous distention. A prominent bruit is heard over the brachiocephalic arteriovenous fistula. There are crackles heard at both lung bases. Cardiac examination shows an S3 gallop. The abdomen is soft and nontender.

Case problem :

Which of the following is the most likely cause of this patient's symptoms?

A. Arteriovenous fistula aneurysm

B. Dialysis disequilibrium syndrome

C. Sepsis

D. Pulmonary embolism

E. Constrictive pericarditis

F. High-output heart failure

Heart Failure (HF)

A clinical syndrome in which the heart can't supply enough blood to meet the body's demands

Caused by structural or functional impairment of ventricular filling or ejection of blood

Leads to the cardinal clinical symptoms of dyspnea, fatigue, and signs of HF like edema and rales

Classifications:

- Heart failure with reduced ejection fraction (HFrEF, systolic HF)

  Vs.

  Heart failure with preserved ejection fraction (HFpEF, diastolic HF)

- Right heart failure (RHF)

Vs. Left heart failure (LHF)

Vs. Biventricular

- Chronic compensated HF

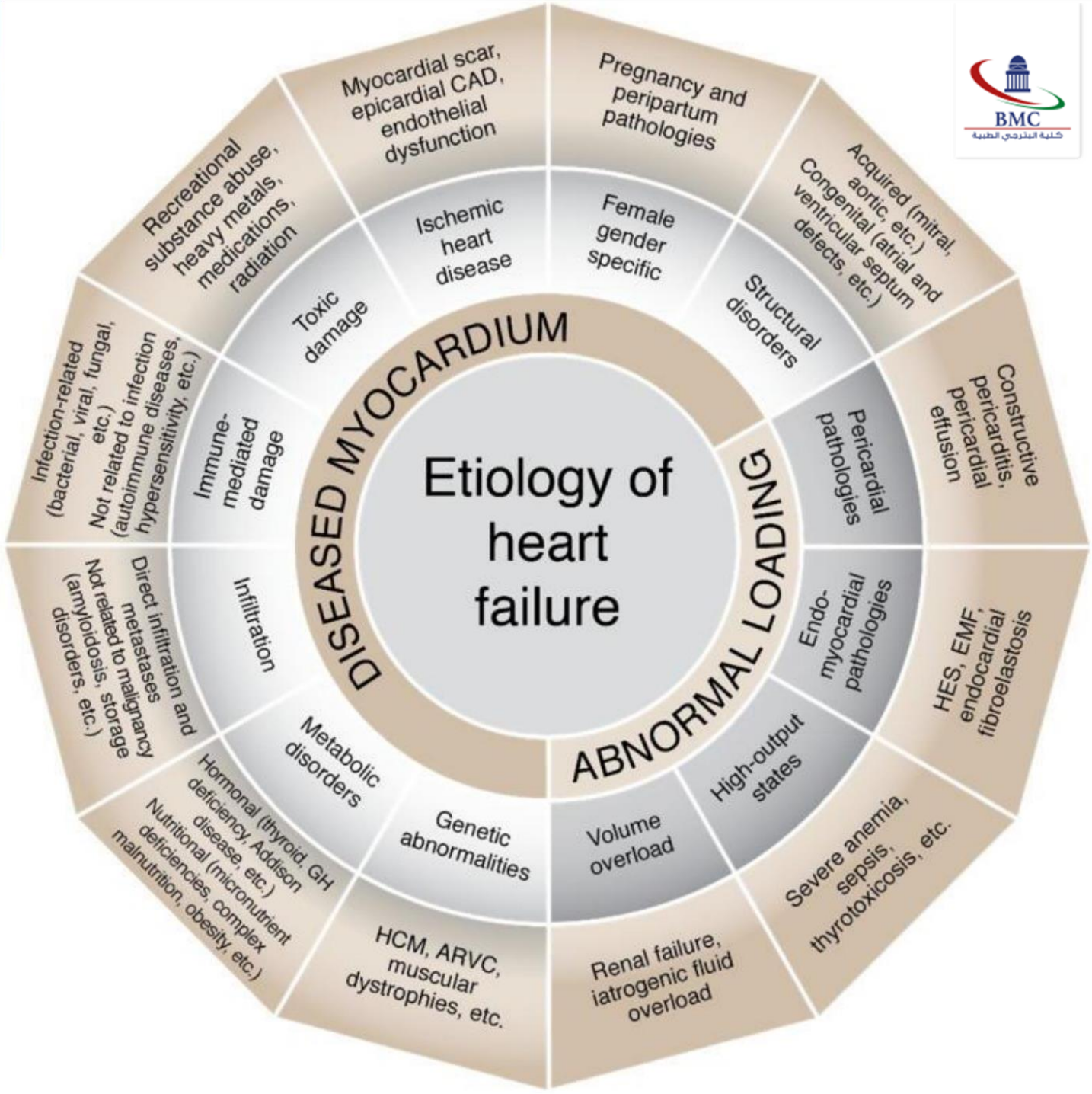
Vs. Acute decompensated H

Epidemiology

- A leading cause of morbidity and mortality worldwide
- Prevalence ~ 2% in adults from developed countries
- Increases ≥ 10% in those aged over 70
- The mean age of patients with HF in KSA is 57-60 years
  (10 years younger than in developed countries)
- Incidence of HF is lower in women than men
- Due to women's longer life expectancy, they still account for half the cases of HF

Etiology

- **Coronary artery disease**
- **Arterial hypertension**
- **Diabetes mellitus**
- Valvular heart disease
- Renal disease
- Infiltrative diseases
- Dilated cardiomyopathy
- Cardiac arrhythmias
- Myocarditis
- Obesity
- Smoking
- COPD
- Drugs
- Alcohol abuse



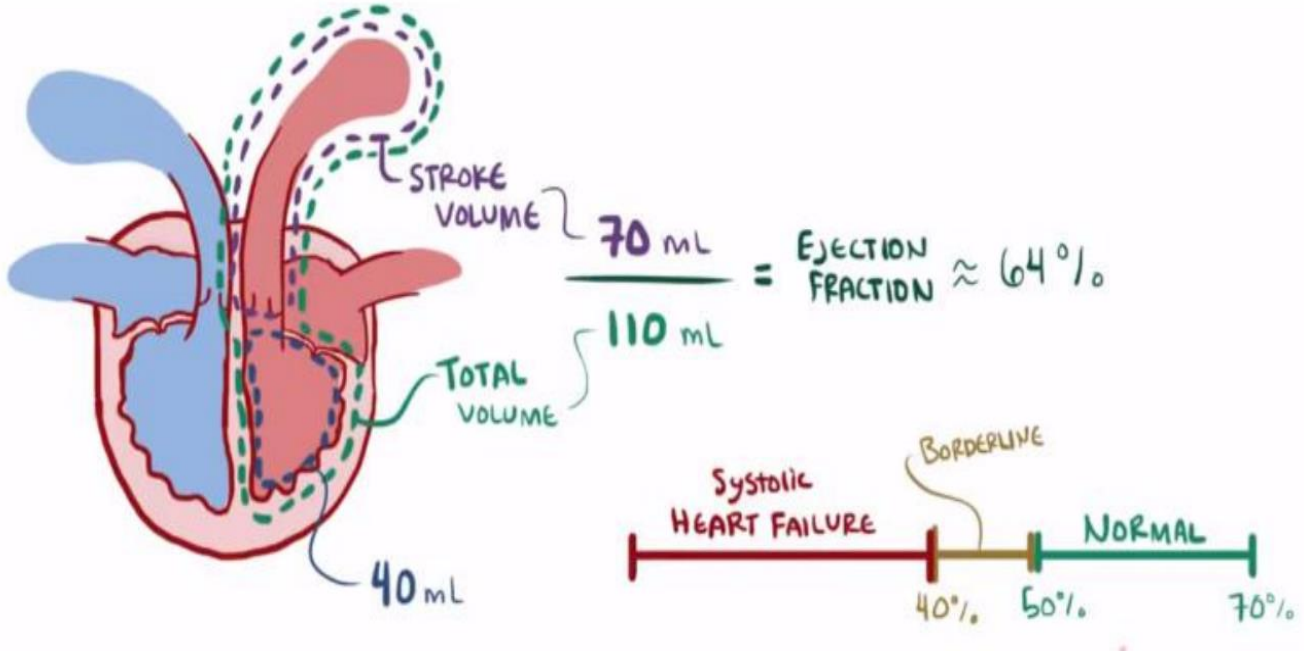
Pathophysiology

- **Cardiac output** is the amount of blood ejected by the heart (L/min)
- CO = HR x SV = $\left[\frac{\text{BEATS}}{\text{MINUTE}}\right] \times \left[\frac{\text{VOLUME BLOOD}}{\text{BEAT}}\right]$

$$70 \frac{\text{beats}}{\text{min}} \times 70 \frac{\text{mL}}{\text{beat}} = 4900 \frac{\text{mL}}{\text{min}}$$

$$\approx 5 \frac{\text{L}}{\text{min}}$$

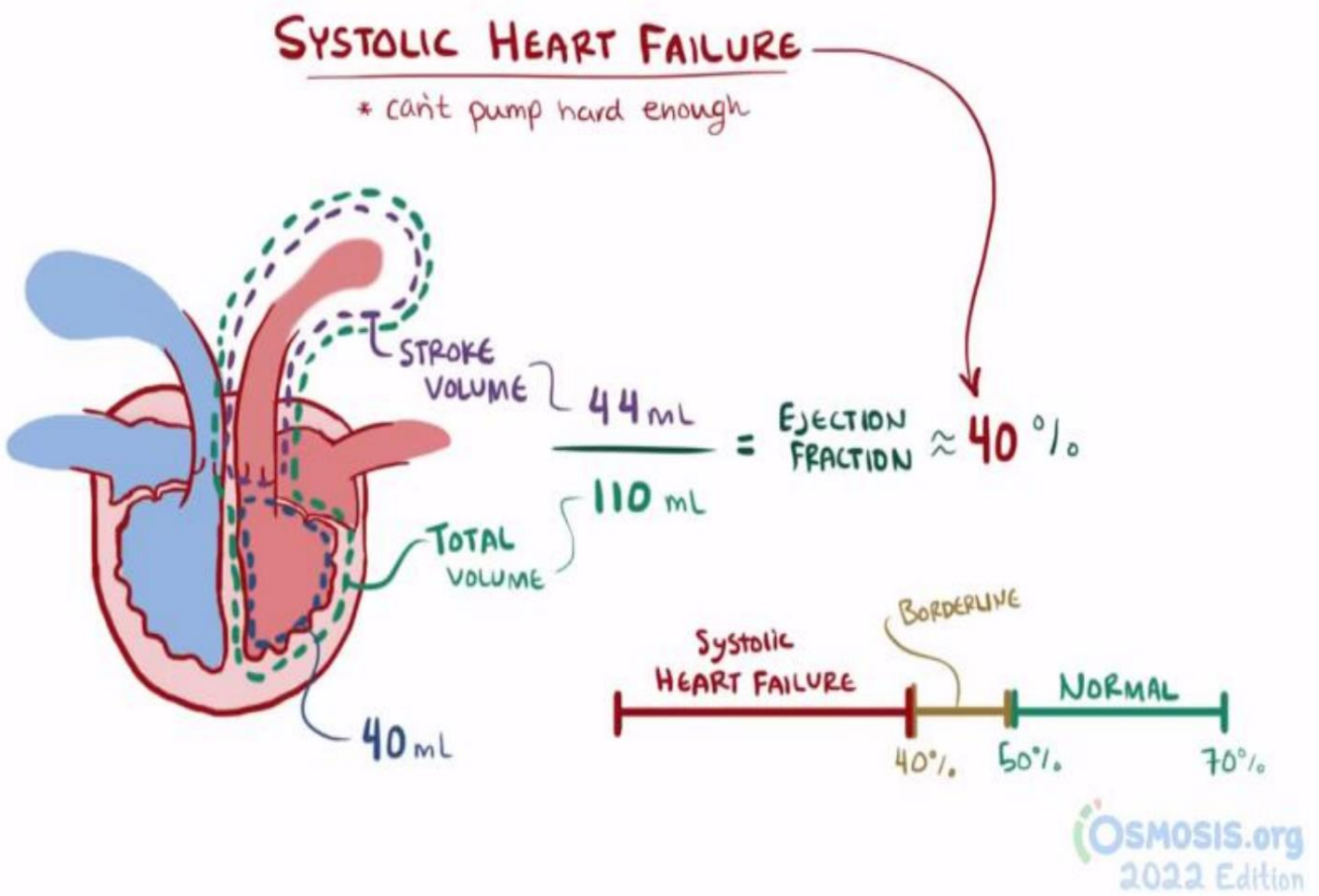
- **Ejection fraction** (EF) is a percentage of how much blood the left ventricle pumps out with each contraction out of the total volume



Pathophysiology

Heart failure with reduced ejection fraction (HFrEF)

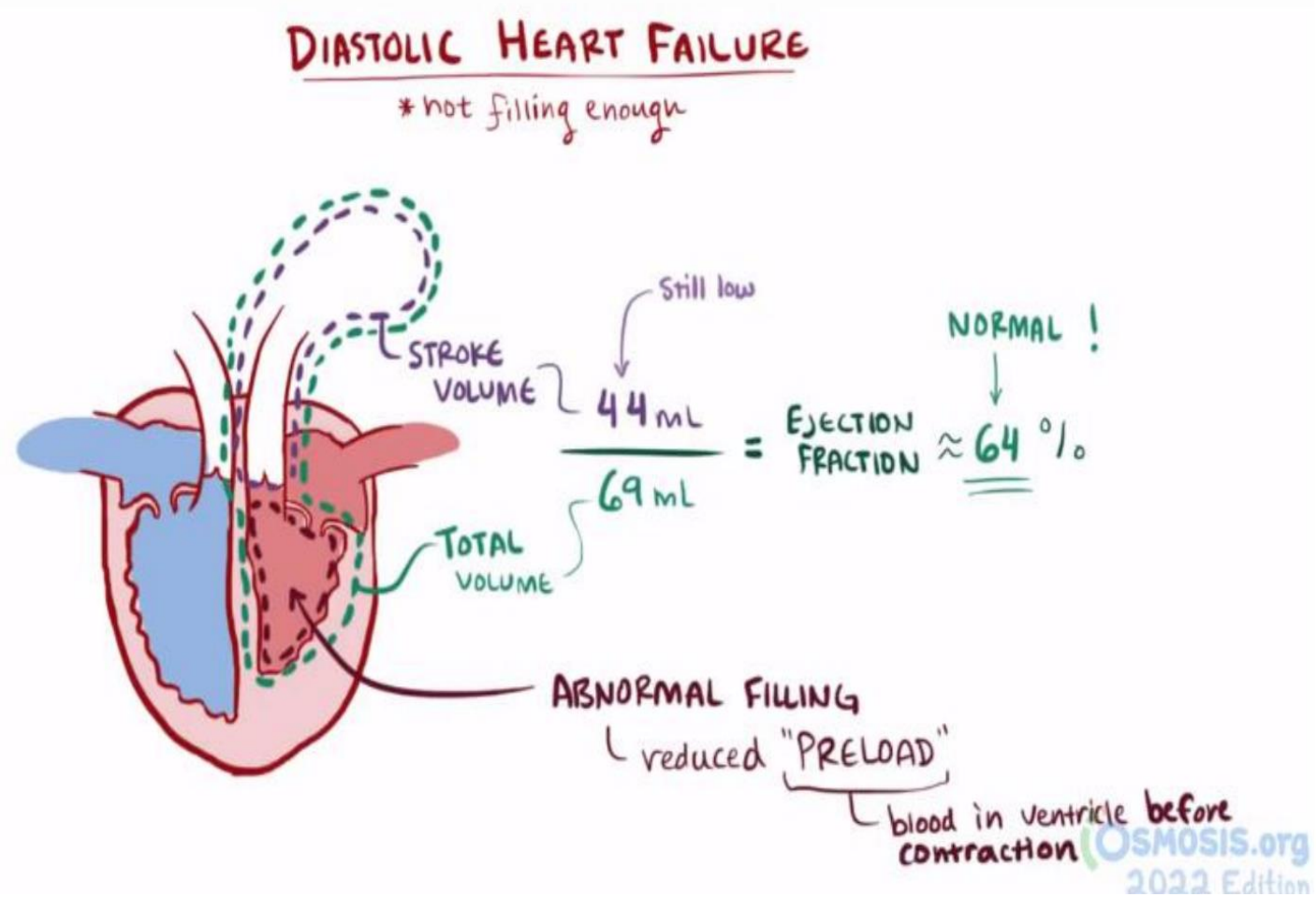
- Reduced contractility → systolic ventricular dysfunction → decreased left ventricular ejection fraction (LVEF) → decreased cardiac output



Pathophysiology

Heart failure with preserved ejection fraction (HFpEF)

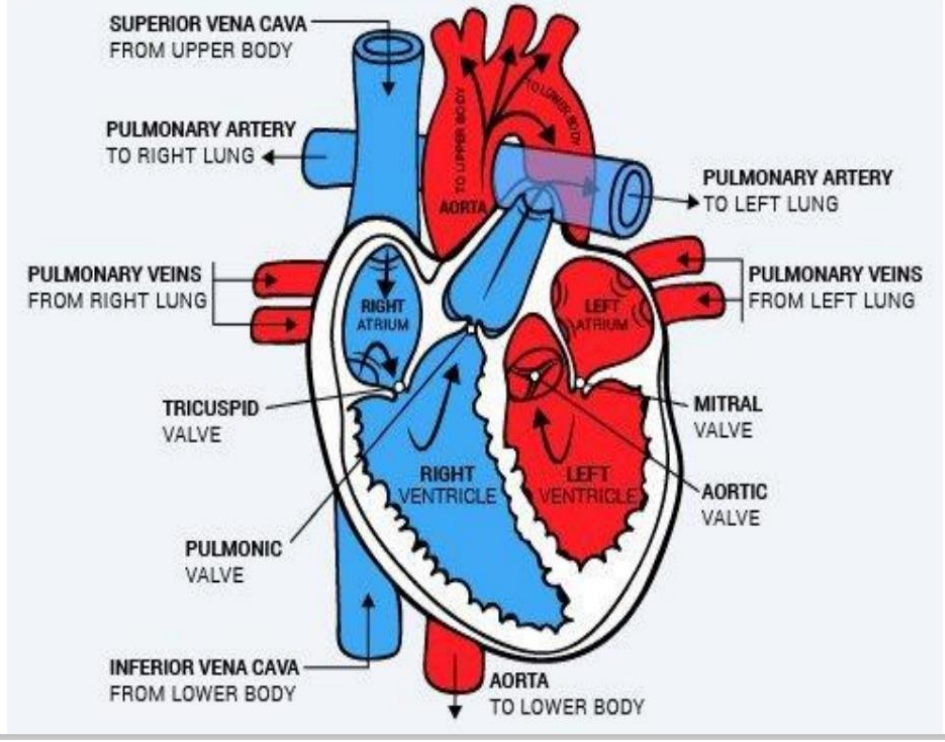
- Decreased ventricular compliance → reduced ventricular filling & increased diastolic pressure → decreased cardiac output (normal EF)



Consequences of Decompensated Heart Failure

- **Forward failure**: reduced cardiac output → poor organ perfusion → organ dysfunction

- **Backward failure**
  - Left-ventricular failure → backup of blood into lungs
  - Right-ventricular failure → systemic venous congestion



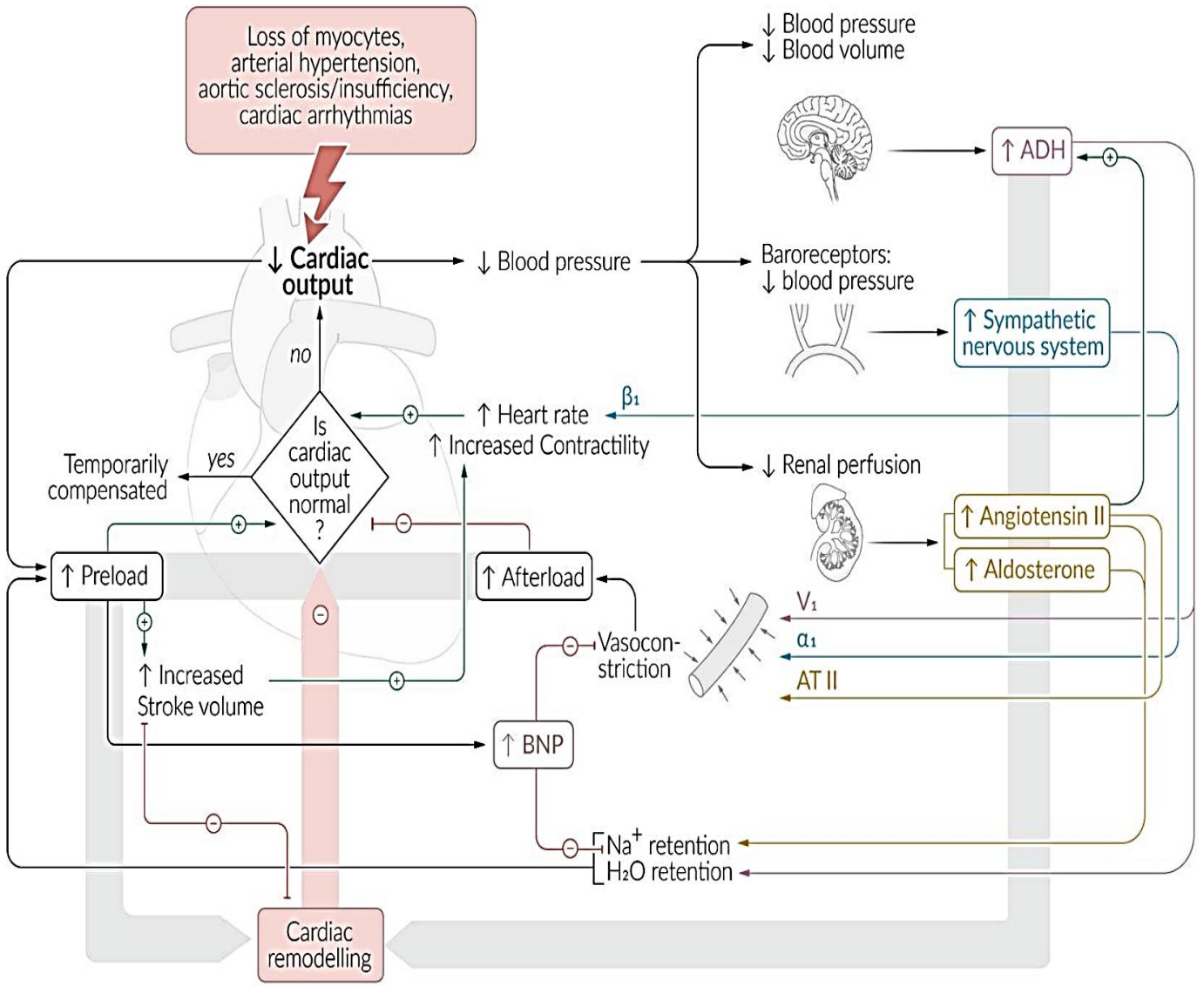
Compensation Mechanisms

To maintain the cardiac output when stroke volume is reduced

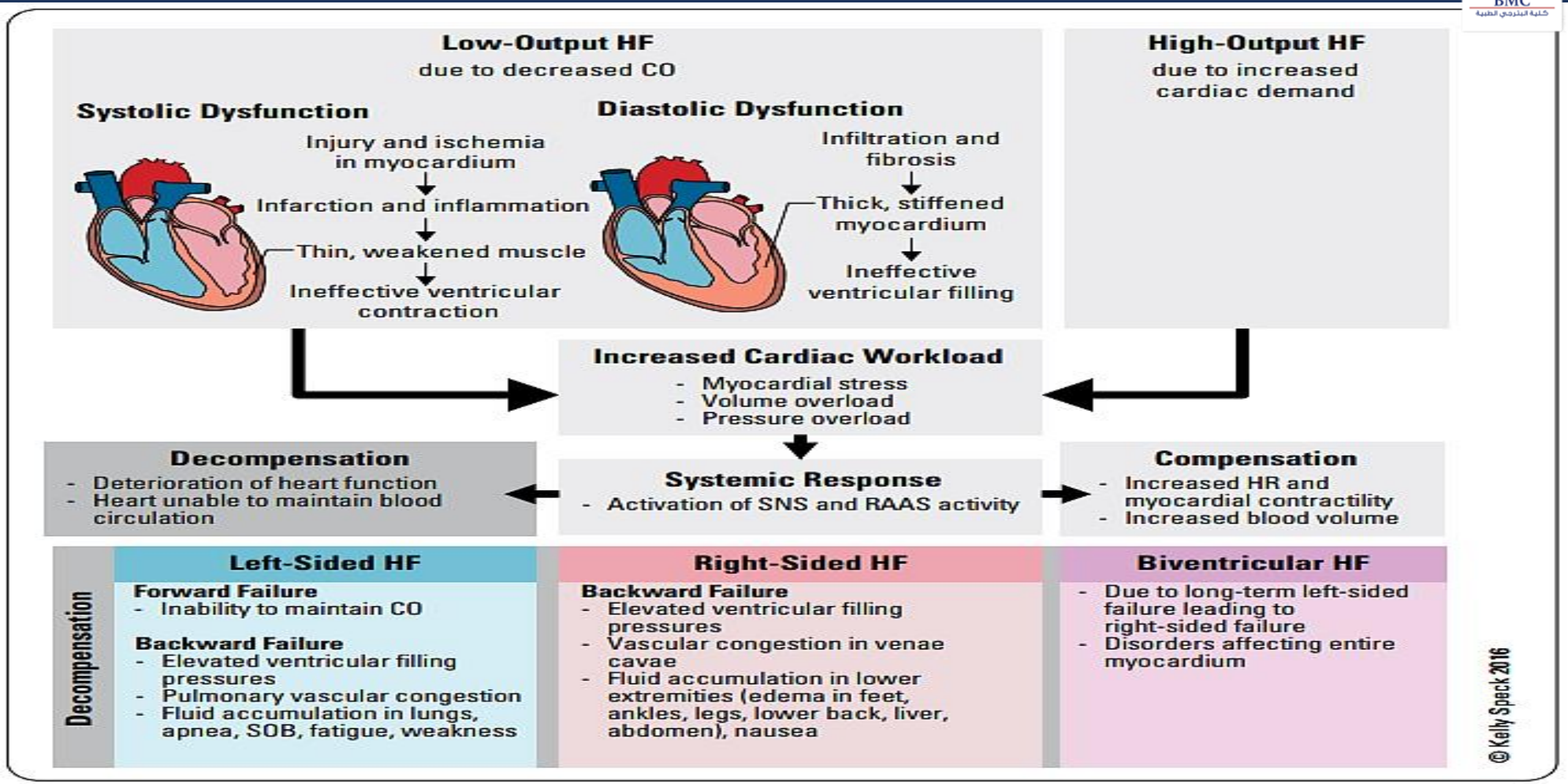
- Increased adrenergic activity

- Increase of renin - angiotensin-aldosterone system activity (RAAS)

- Secretion of brain natriuretic peptide (BNP)

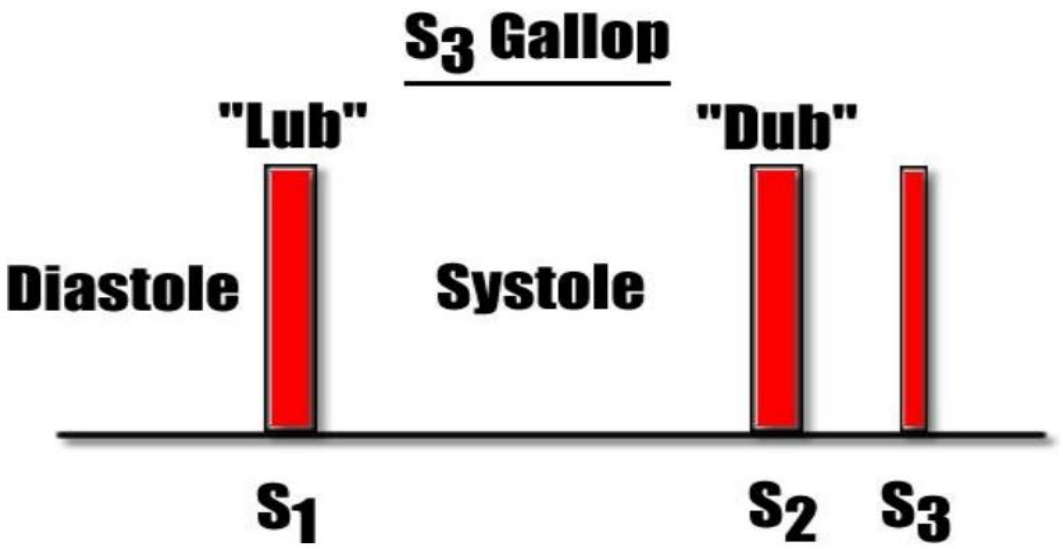


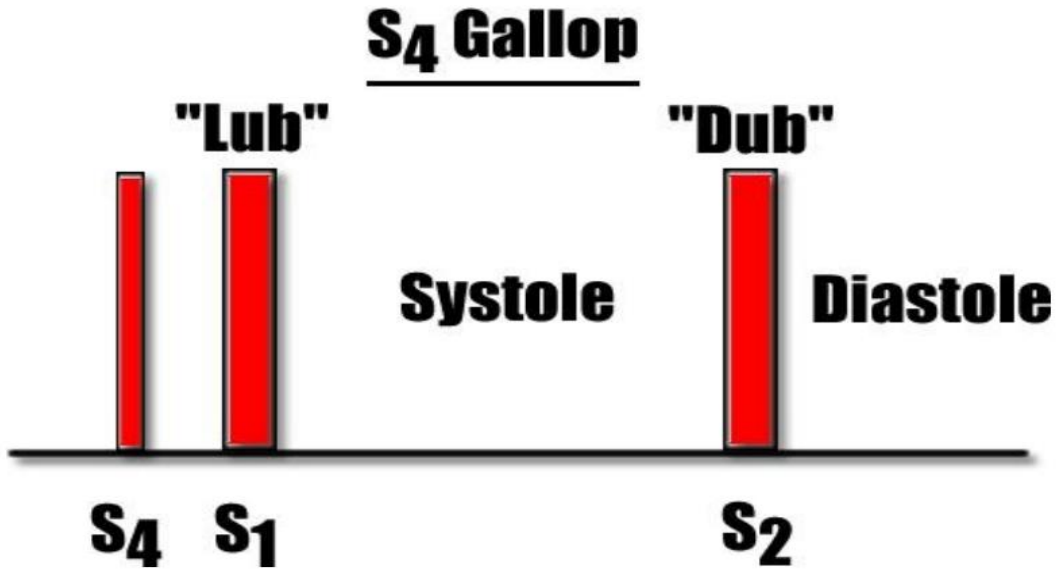
Decompensation Effects



General Clinical Features

- Nocturia
- Fatigue
- Tachycardia
- Various arrhythmias
- S3 gallop
- S4 gallop
- Cachexia





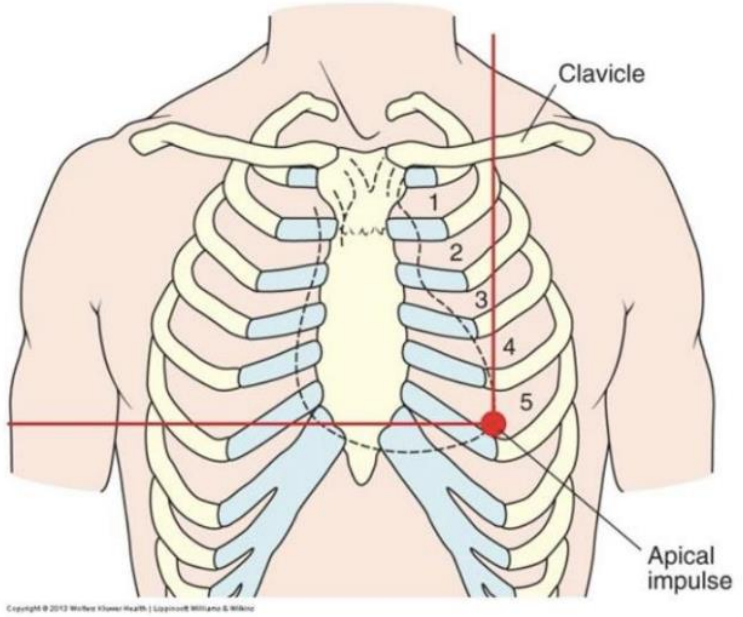
Clinical Features – Left-sided HF

Symptoms of pulmonary congestion

- Dyspnea
- Orthopnea
- Pulmonary edema
- Paroxysmal nocturnal dyspnea

Physical examination findings

- Bilateral basilar rales
- Laterally displaced apical heartbeat
- Coolness and pallor of lower extremities



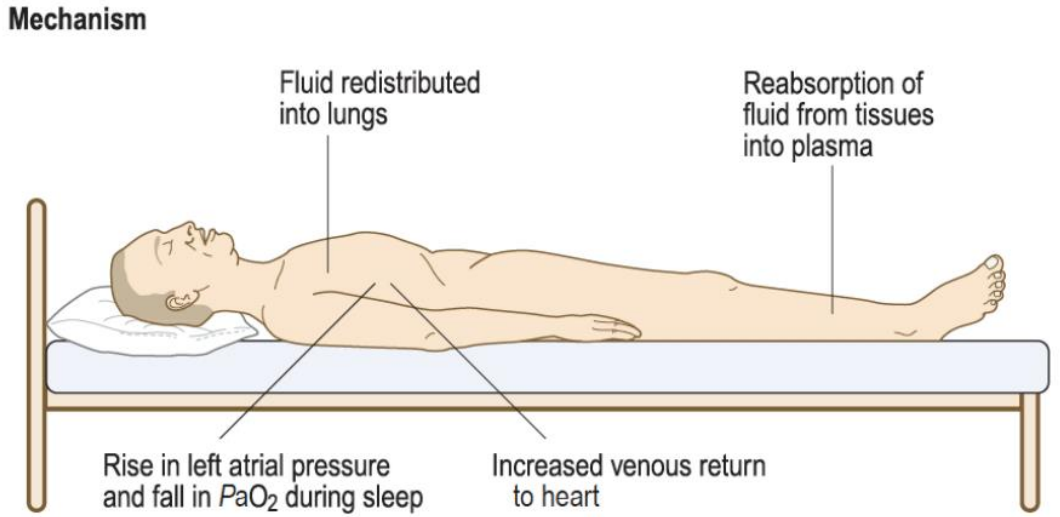


Fig. 4.3 Paroxysmal nocturnal dyspnoea.

Clinical Features – Right-sided HF

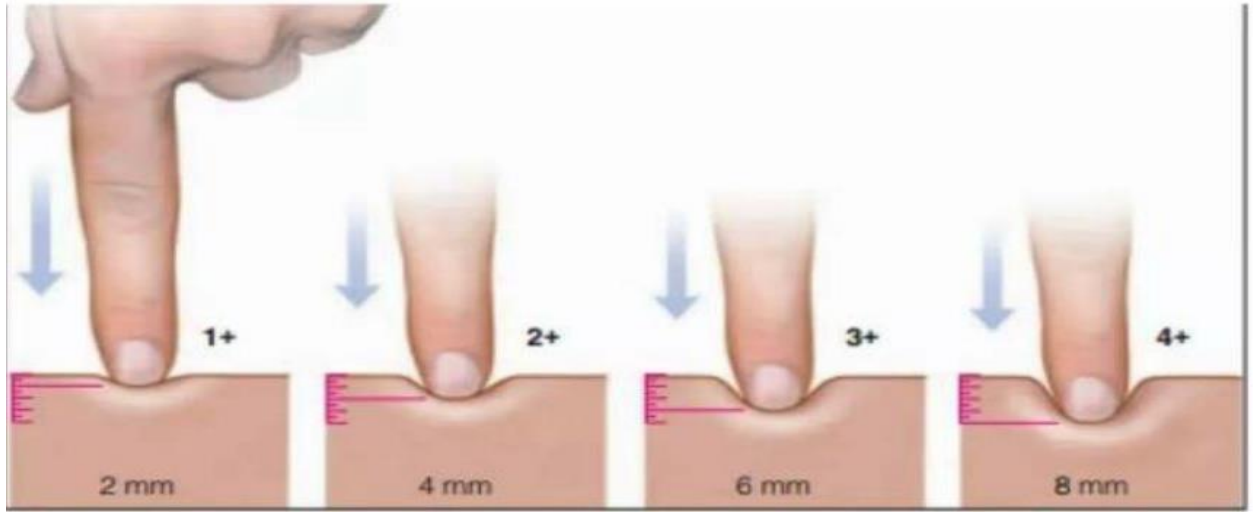
Symptoms of fluid retention and increased CVP

- Peripheral pitting edema
- Abdominal pain
- Jaundice
- Nausea
- Loss of appetite

1+	2mm depression, barely detectable. Immediate rebound.
2+	4mm deep pit. A few seconds to rebound.
3+	6mm deep pit. 10-12 seconds to rebound.
4+	8mm: very deep pit. >20 seconds to rebound.

Physical examination findings

- Jugular venous distention
- Hepatojugular reflux
- Hepatosplenomegaly



The ACC/AHA Stages of Heart Failure



4.5 New York Heart Association classification of heart failure symptom severity

Class	Description
I	No limitations. Ordinary physical activity does not cause undue fatigue, dyspnoea or palpitation (asymptomatic left ventricular dysfunction)
II	Slight limitation of physical activity. Such patients are comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnoea or angina pectoris (symptomatically 'mild' heart failure)
III	Marked limitation of physical activity. Less than ordinary physical activity will lead to symptoms (symptomatically 'moderate' heart failure)
IV	Symptoms of congestive heart failure are present, even at rest. With any physical activity, increased discomfort is experienced (symptomatically 'severe' heart failure)

Diagnostics – Blood work

Everyone:

- CBC
- BMP (↑ Creatinine, ↓ Na)
- Cardiac biomarkers
- Glucose or HbA1c
- Liver chemistries
- Lipid profile

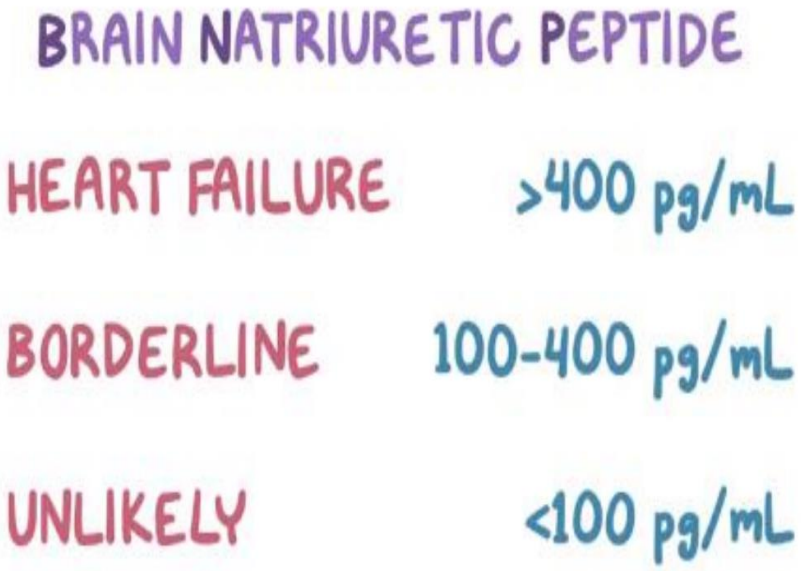
Additional tests for specific populations:

- HIV
- Iron studies
- ANA
- Lyme serology
- TSH

Diagnostics – Cardiac Biomarkers

Natriuretic peptides (NP): BNP or NT-proBNP

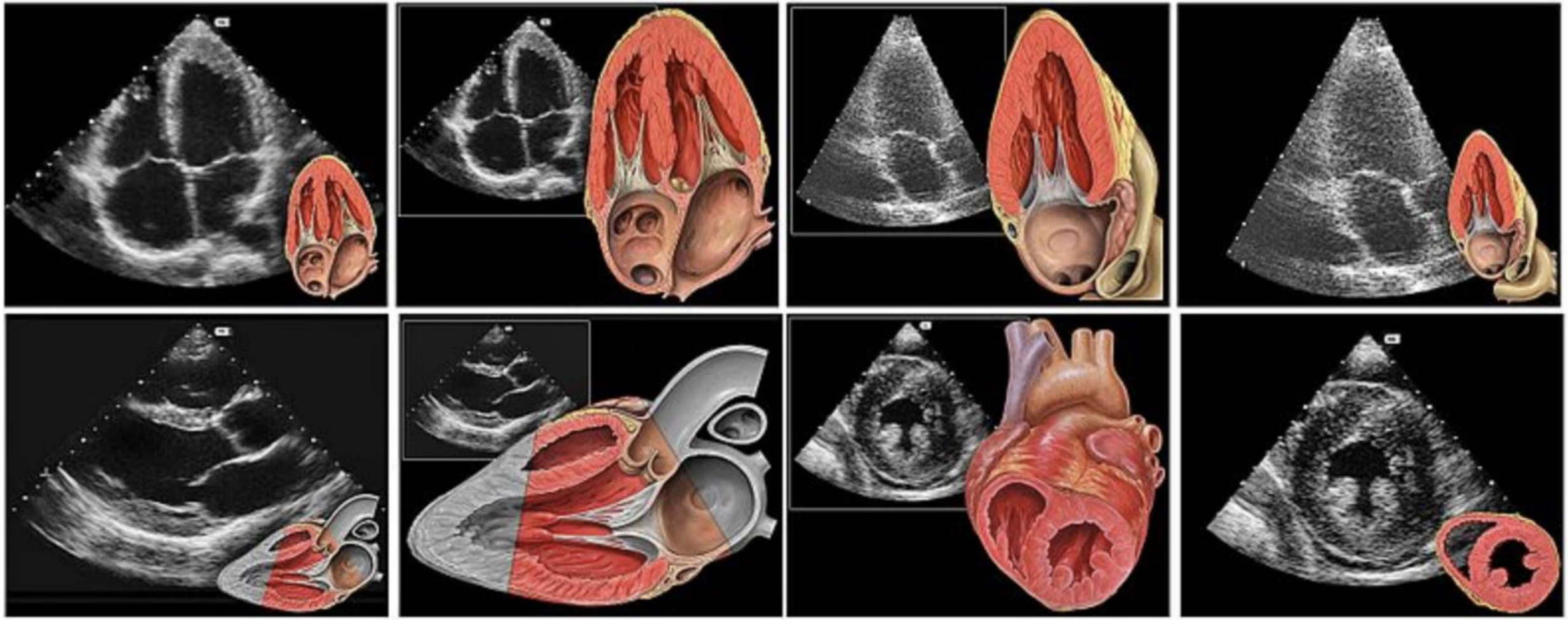
- Released in response to ventricular volume expansion & pressure overload
- Useful in differentiating between dyspnea caused by CHF and COPD
- Can be elevated with age and kidney dysfunction
- Falsely decreased with obesity



Diagnostics – Imaging Studies

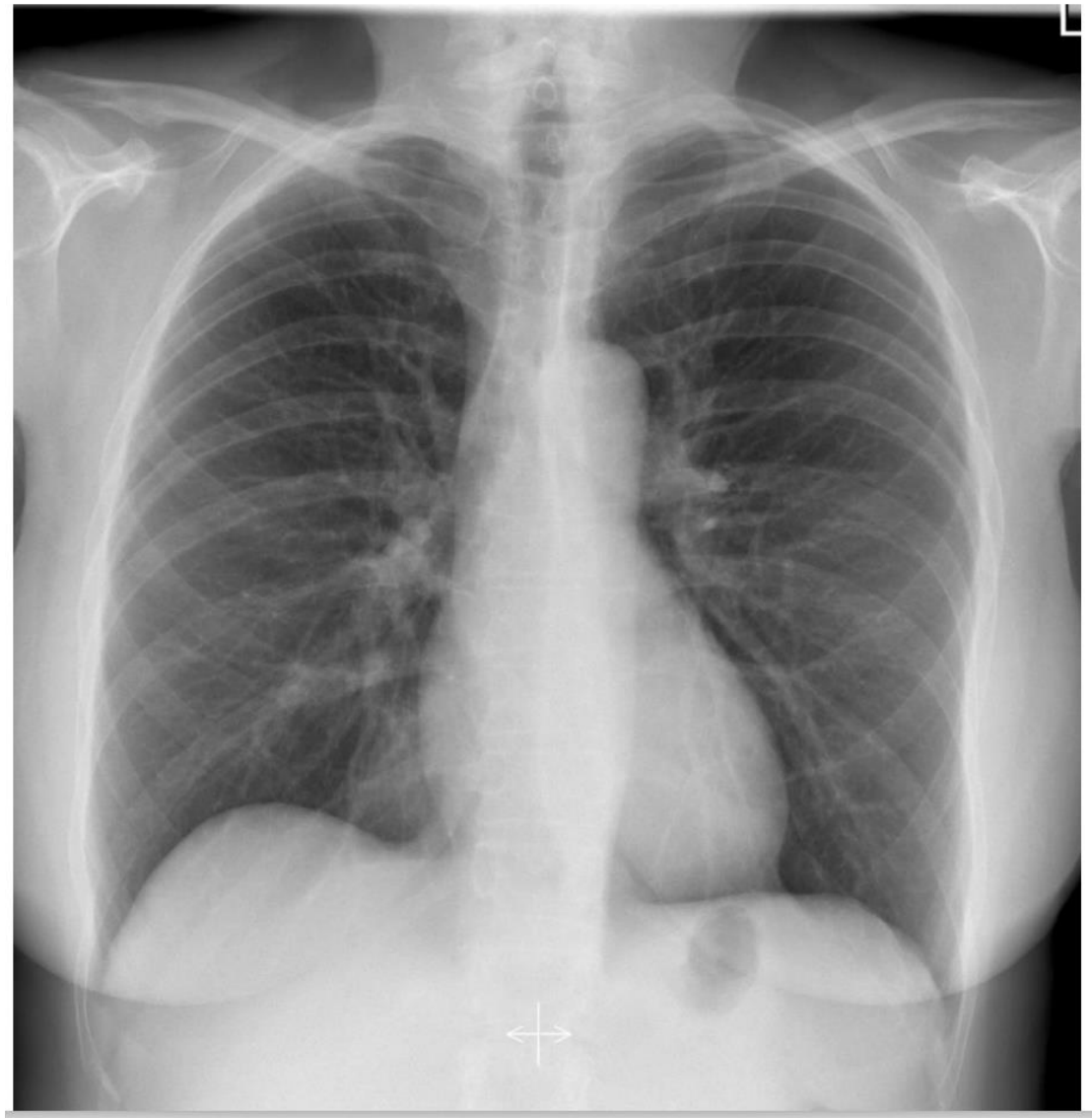
Transthoracic echocardiogram (TTE)

- Indicated in Every patient with suspected new-onset HF



Diagnostics – Imaging Studies

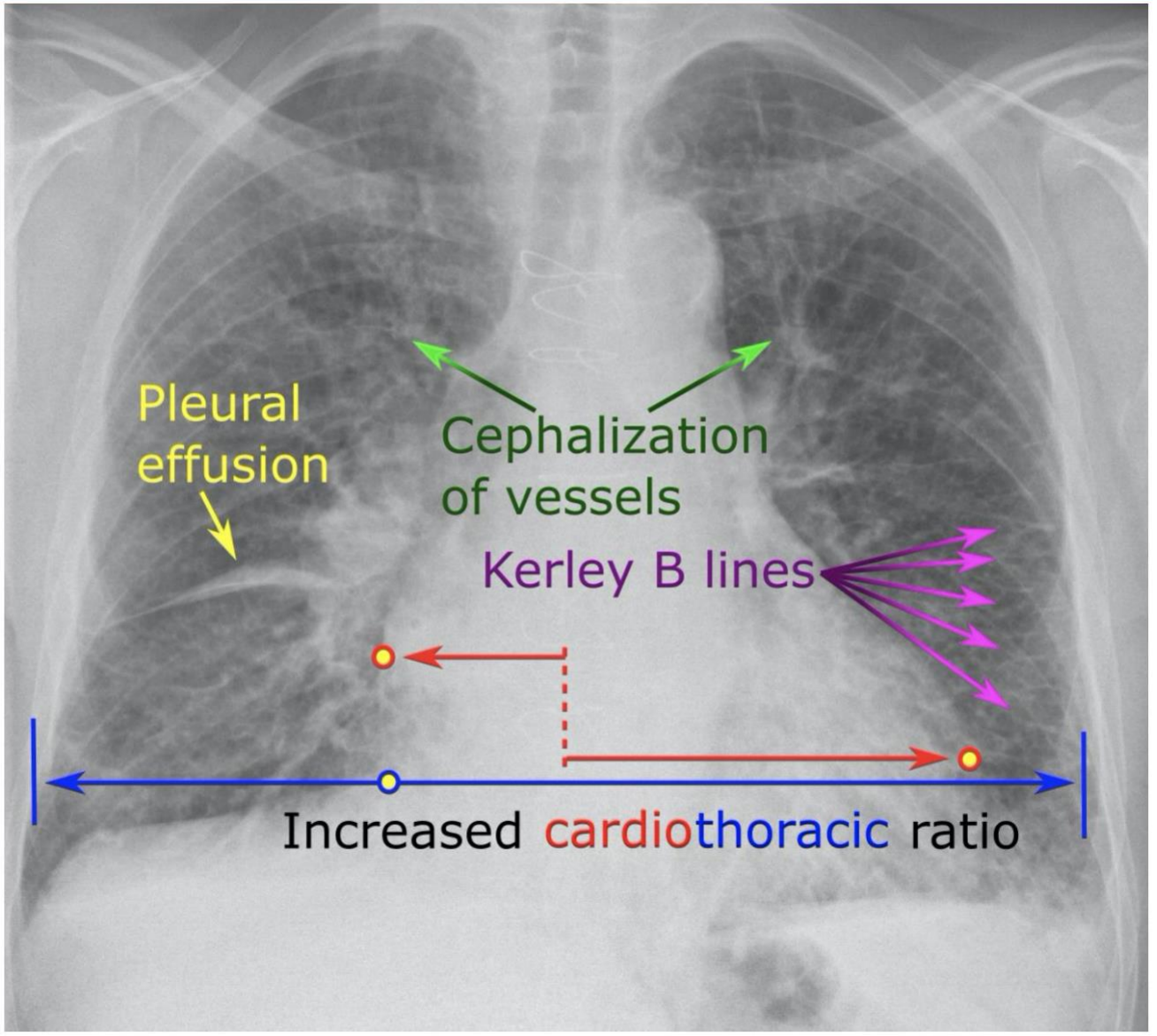
Chest x-ray





Diagnostics – Imaging Studies

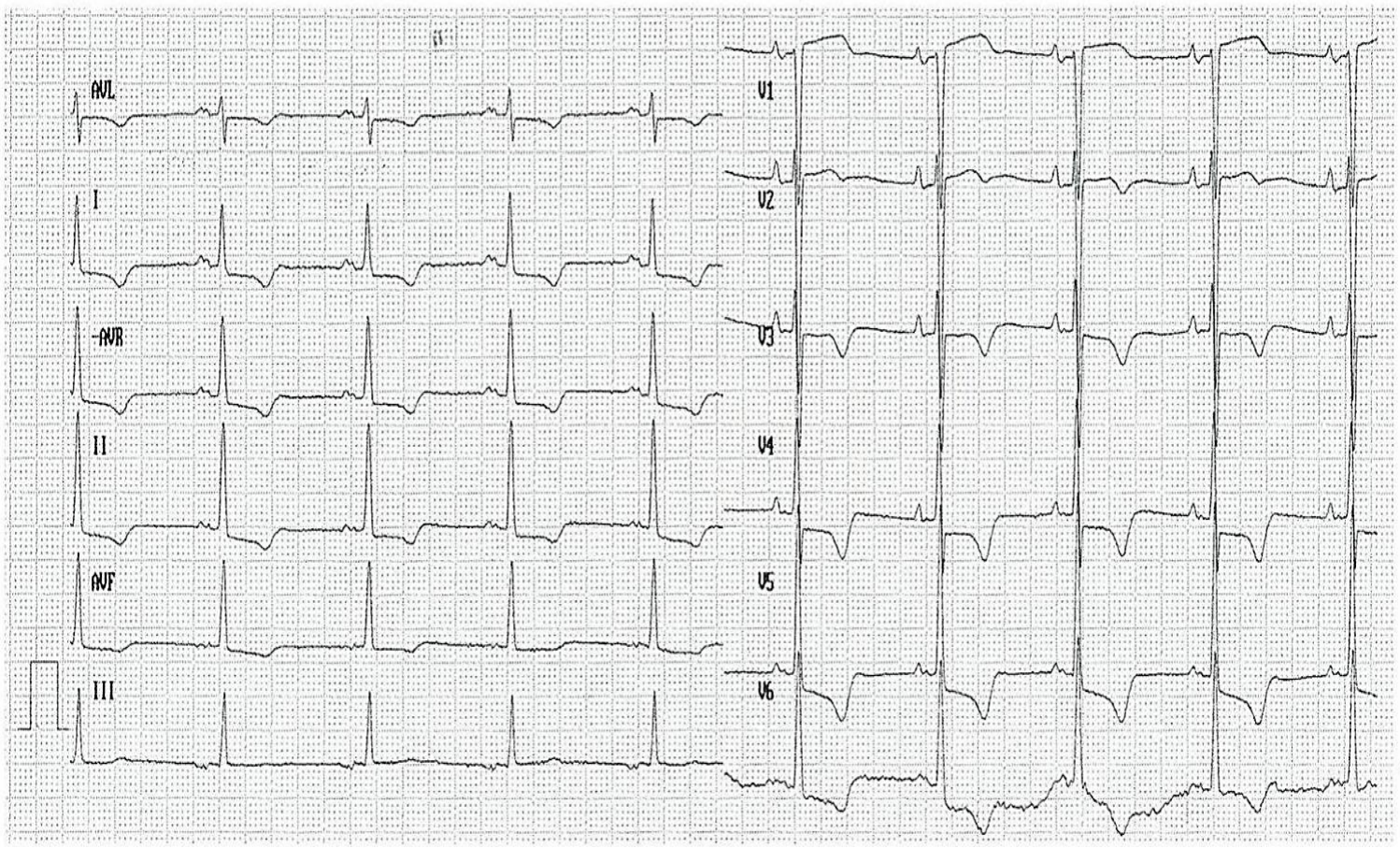
Chest x-ray

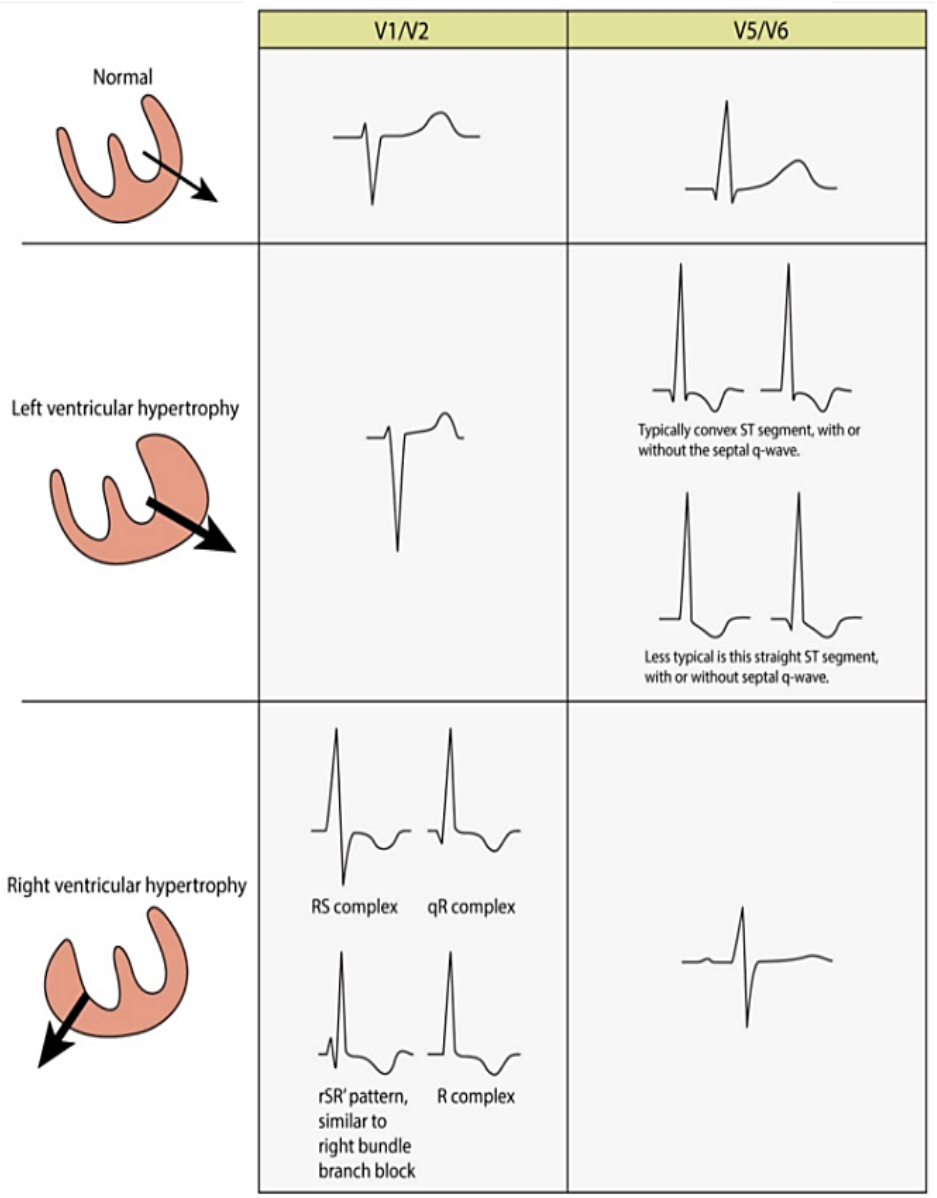


Diagnostics – Imaging Studies

Electrocardiogram (ECG)

- Can detect chamber enlargement
- Can detect presence of ischemic heart disease



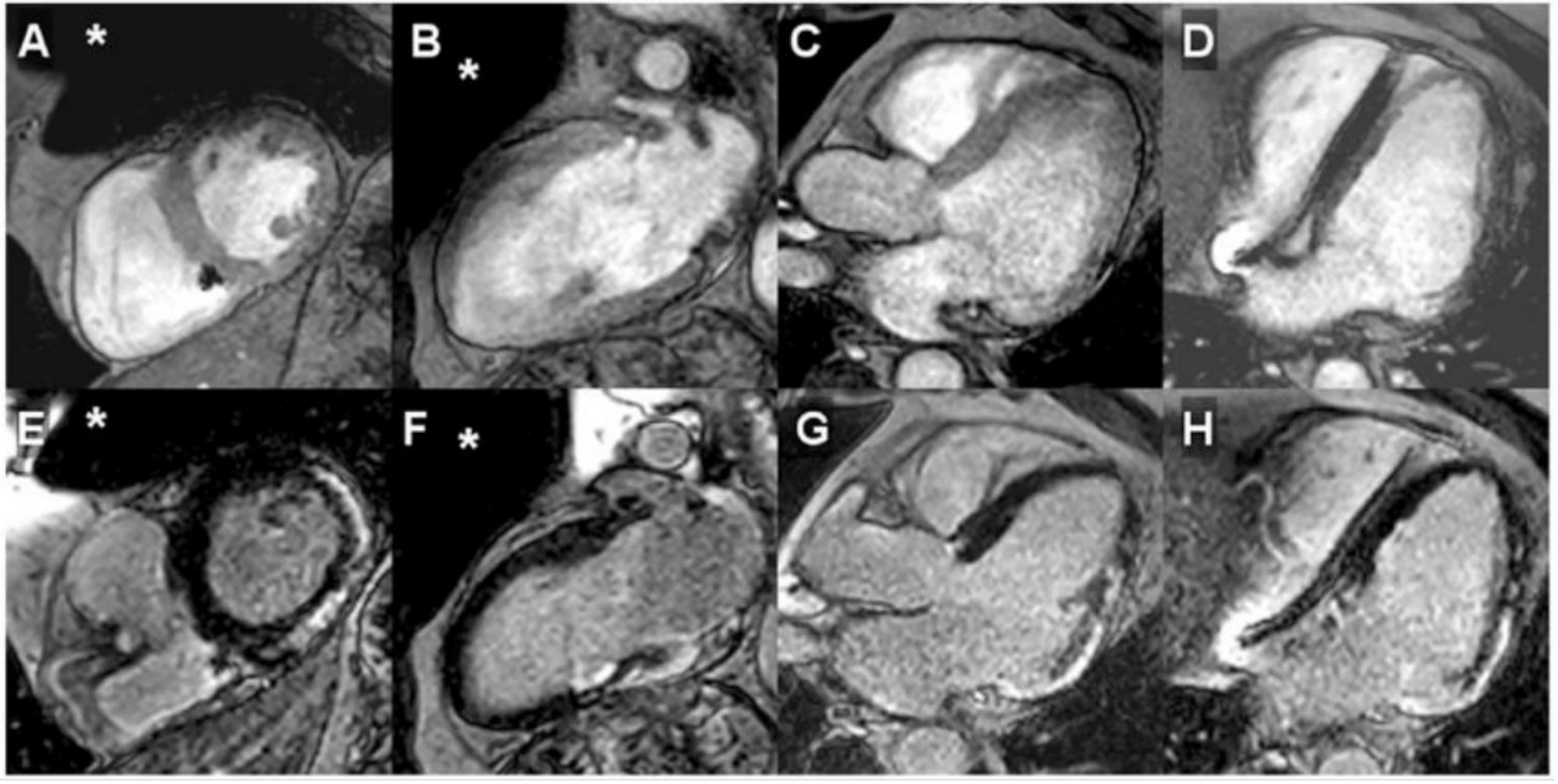


Diagnostics – Imaging Studies

Cardiac MRI

Gold standard for assessment of ventricular volume, mass, and EF

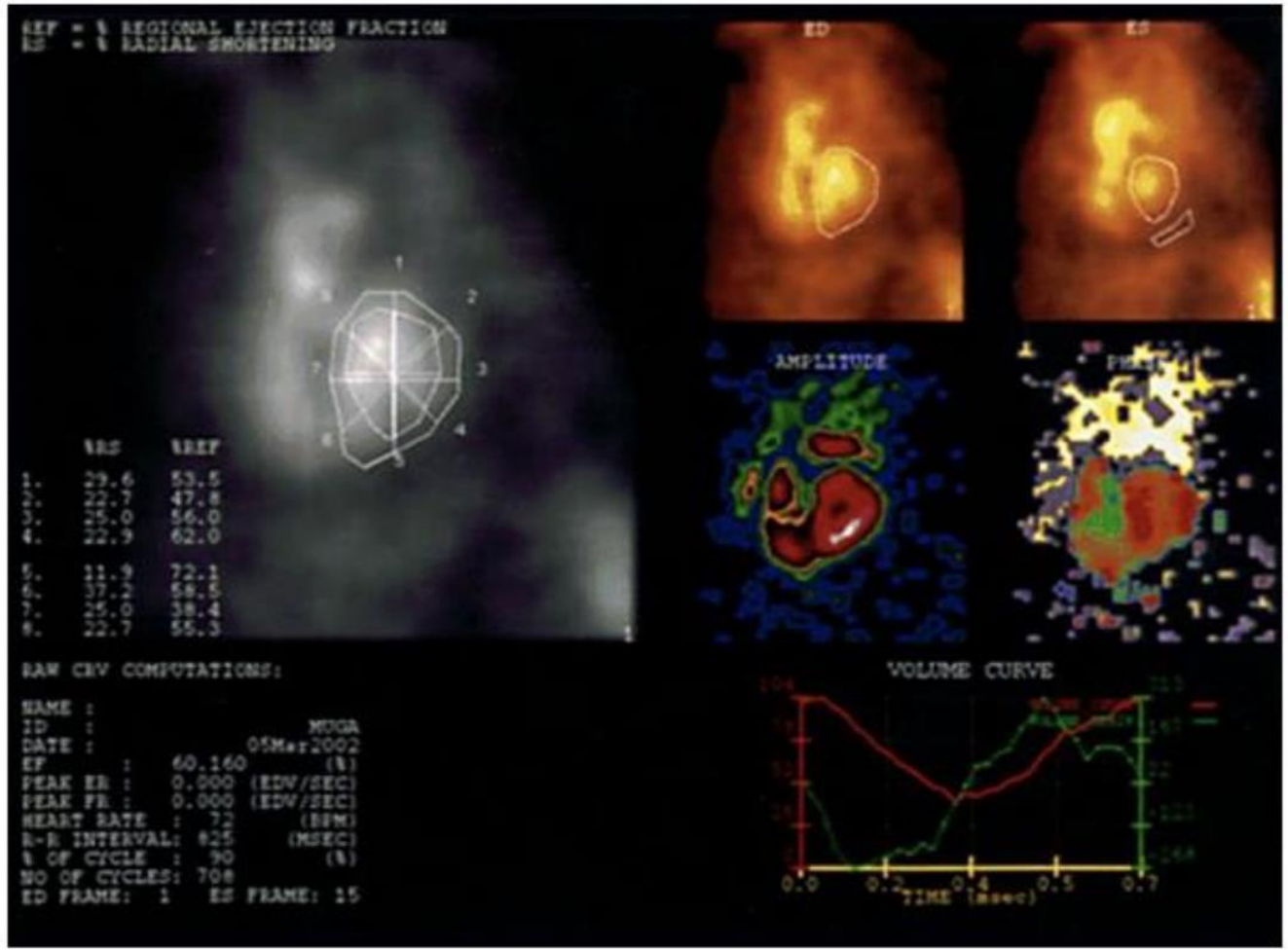
- Indicated when there's diagnostic uncertainty after a TTE or investigating certain diseases



Diagnostics – Imaging Studies

Radionucleotide ventriculography

- Also called a multi-gated acquisition scan or MUGA scan
- Used to assess EF and volume when other modalities are inadequate or contraindicated



Diagnosis of congestive heart failure

- **<u>Framingham criteria for diagnosis of congestive heart failure:</u>**

- Diagnosis of CHF requires the presence of at least 2 major criteria or 1 major criterion in conjunction with 2 minor criteria.

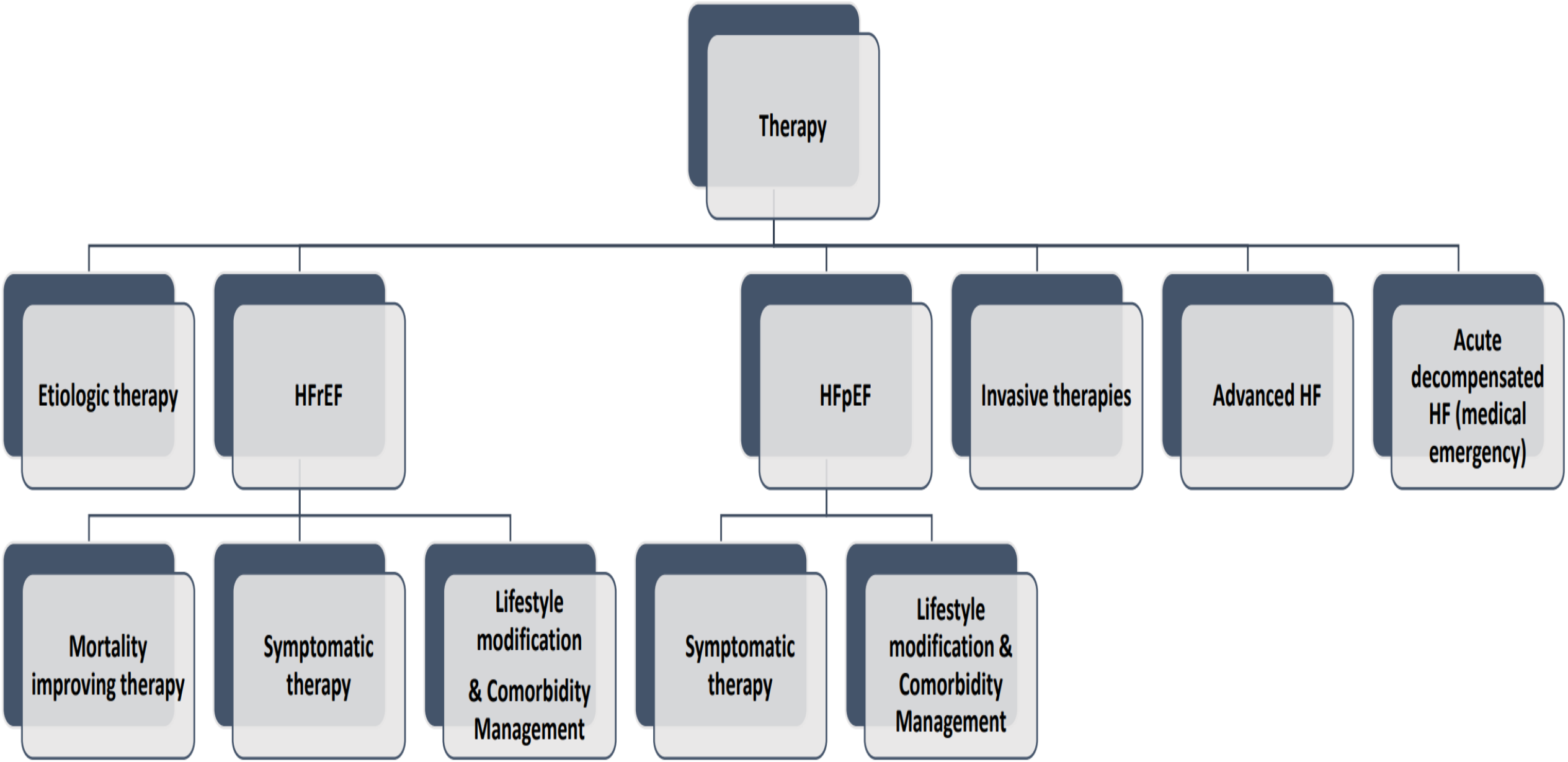
Major criteria :

- Paroxysmal nocturnal dyspnea
- Neck vein distention
- Rales
- Radiographic cardiomegaly.
- Acute pulmonary edema
- S_3 gallop.
- Increased central venous pressure (>16 cm H_2O at right atrium)
- Hepatojugular reflux
- Weight loss >4.5 kg in 5 days in response to treatment.

Minor criteria :

- Bilateral ankle edema
- Nocturnal cough
- Dyspnea on ordinary exertion
- Hepatomegaly
- Pleural effusion
- Decrease in vital capacity by one third from maximum recorded
- Tachycardia (heart rate>120 beats/min.)

Management


Therapy
Etiologic therapy
HFrEF
HFpEF
Invasive therapies
Advanced HF
Acute decompensated HF (medical emergency)
Mortality improving therapy
Symptomatic therapy
Lifestyle modification & Comorbidity Management
Symptomatic therapy
Lifestyle modification & Comorbidity Management

Management – Lifestyle Modifications

- Sodium restriction (less than 4 g/day)
- Fluid restriction (1.5 to 2.0 L daily)
- Weight loss
- Smoking cessation
- Restrict alcohol use
- Exercise program
- Daily weight monitoring
- Immunization
- Comorbidity management
- Avoidance of medications that may worsen HF

Management -HFrEF

Initial medical treatment for heart failure

Stage A

Treatment of cardiovascular risk factors	Treat hypertension, hyperlipidemia, diabetes if present. HF medications not routinely recommended

Stage B

Class	Example	Indications	Medication effect
ACE inhibitors (ACEi)	Enalapril Ramipril Lisinopril	Everyone with HFrEF	Mortality improving
Angiotensin receptor blockers (ARBs)	Candesartan Losartan Valsartan	Patients who can't tolerate an ACEi	Mortality improving
Angiotensin receptor-neprilysin inhibitors (ARNIs)	sacubitril/valsartan	Patients who can't tolerate an ACEi/ARB	Mortality improving
β-blockers	Bisoprolol Carvedilol Metoprolol	Everyone with HFrEF	Mortality improving

Or

Or

Management -HFrEF

Initial medical treatment for heart failure

Stage C

Class	Example	Indications	Medication effect
Aldosterone antagonists	Spironolactone Eplerenone	NYHA II-IV and an **LVEF ≤ 35%**	Mortality improving
SGLT-2 inhibitor	Dapagliflozin Empagliflozin	NYHA II-IV	Mortality improving
Loop Diuretics	Torsemide Furosemide	All patients with fluid retention	Improves symptoms & reduces hospitalization
Thiazide diuretics	Chlorthalidone Hydrochlorothiazide		
Isosorbide dinitrate (ISDN) & Hydralazine	Combo pill (BiDil) Separate dosing	Angiotensin system blockade intolerance	Mortality improving

Stage D

Additional measures	Most patients require invasive interventions or a change in focus to palliative care

Management -HFrEF

Additional medical treatment for heart failure

Class	Indications	Medication effect
Digoxin	HFrEF with persistent symptoms despite treatment with appropriate first-line medications	Symptomatic treatment & reduces hospitalization
Ivabradine	If all of the following are present: • NYHA class II–III • Stable HFrEF ≤ 35% • Sinus rhythm with HR of ≥ 70 bpm at rest on maximum tolerated beta-blocker dose	Symptomatic treatment

Management of chronic heart failure



Side effects: **digitalis toxicity:** **It has a low therapeutic index so must be monitored**

Precipitating factors increasing toxicity:

(Old age, renal failure, Hypokalemia, Hypomagnesaemia, Hypercalcemia. Thyroid disorders).

Clinical picture of toxicity:-

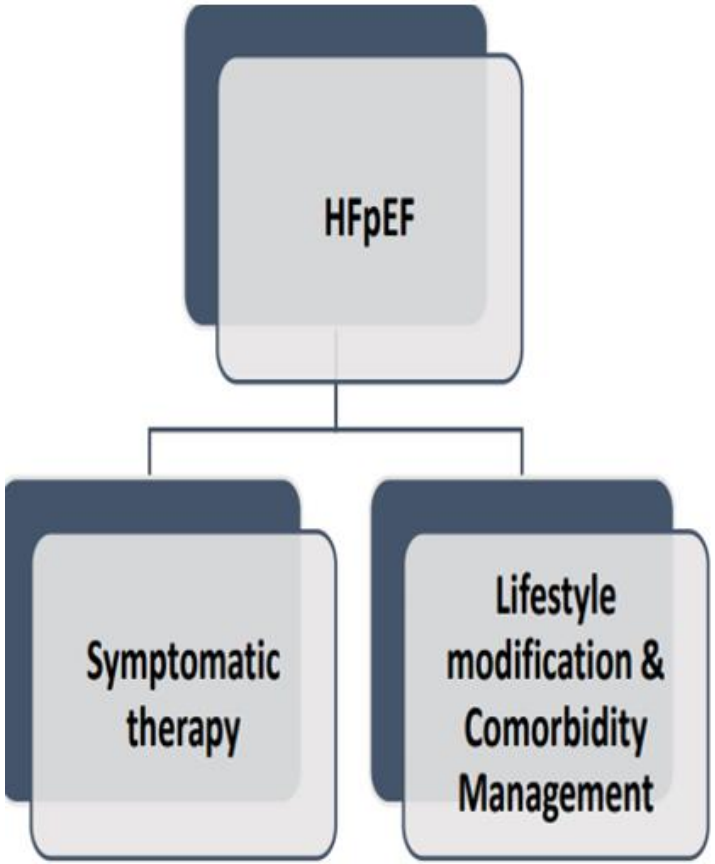
- **Non-cardiac: GIT**: Anorexia, nausea, vomiting (**1st symptom**) → **Neurological**: Psychosis, **yellow vision**. **Cardiac:** (**most life-threatening**): Arrhythmias, (pulses bigeminy and trigeminy), AVN conduction heart block.
- Note: **Any arrhythmia can be a manifestation of digitalis toxicity, except** typeII (mobtiz II) second-degree heart block.

Treatment of toxicity:

- Stop digitalis. → Stop diuretics (especially potassium-losing diuretics) → add Digitalis antibodies (Digibind).
- Anti-arrhythmic drugs (e.g.: phenytoin, lidocaine for ventricular arrhythmias & amiodarone for atrial arrhythmia).
- **To avoid toxicity:** Decrease the dose &**Drug holiday**. Routine **estimation of serum level of digitalis** (N=0.5-2ng/ml).

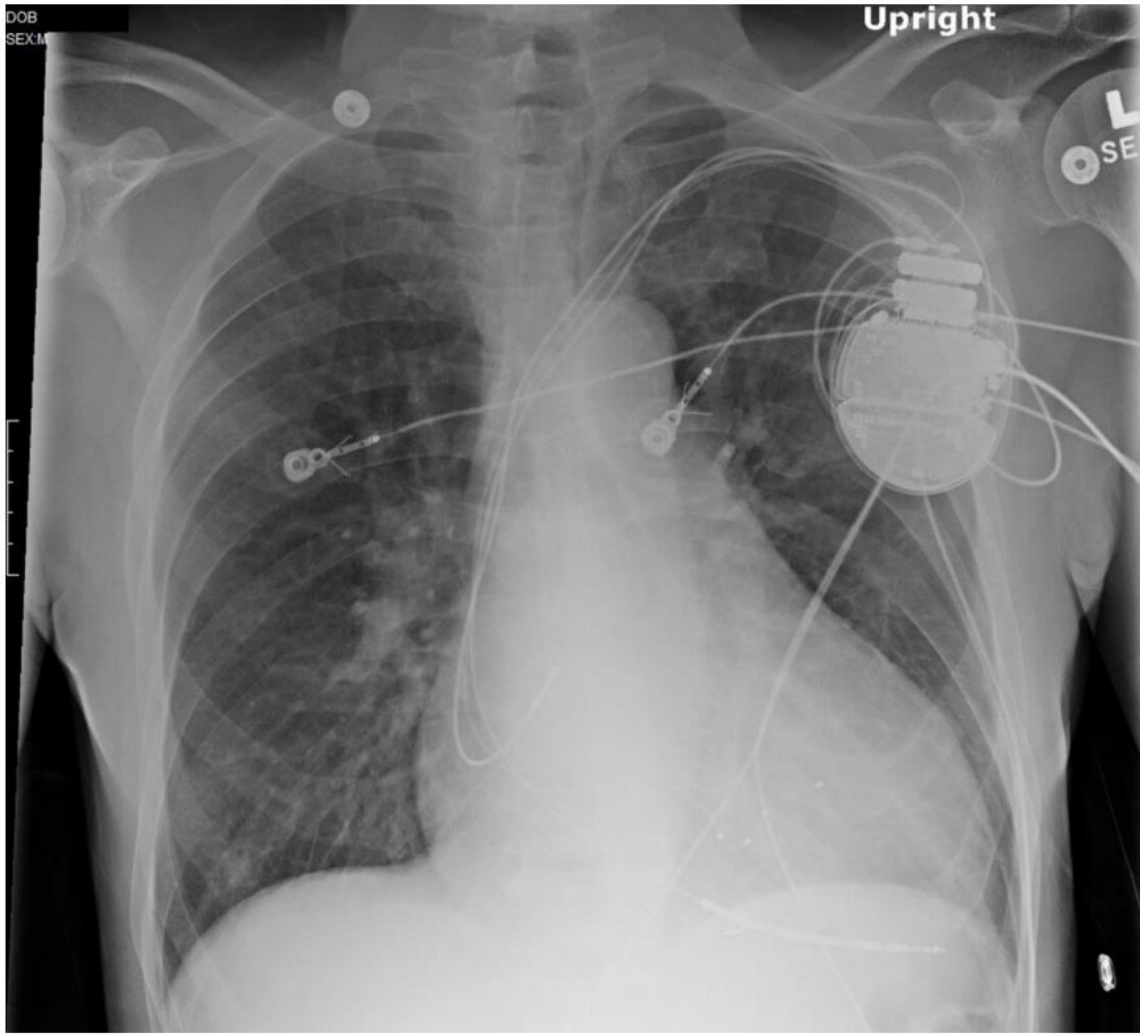
Management - HFpEF

- No therapies with proven mortality benefit
- Diuretics are mainstay for symptom control
- Induction of bradycardia if tachycardia exists and relief of ischemia if present
- Lifestyle modification & comorbidity management same as for HFrEF



Management – Invasive Interventions

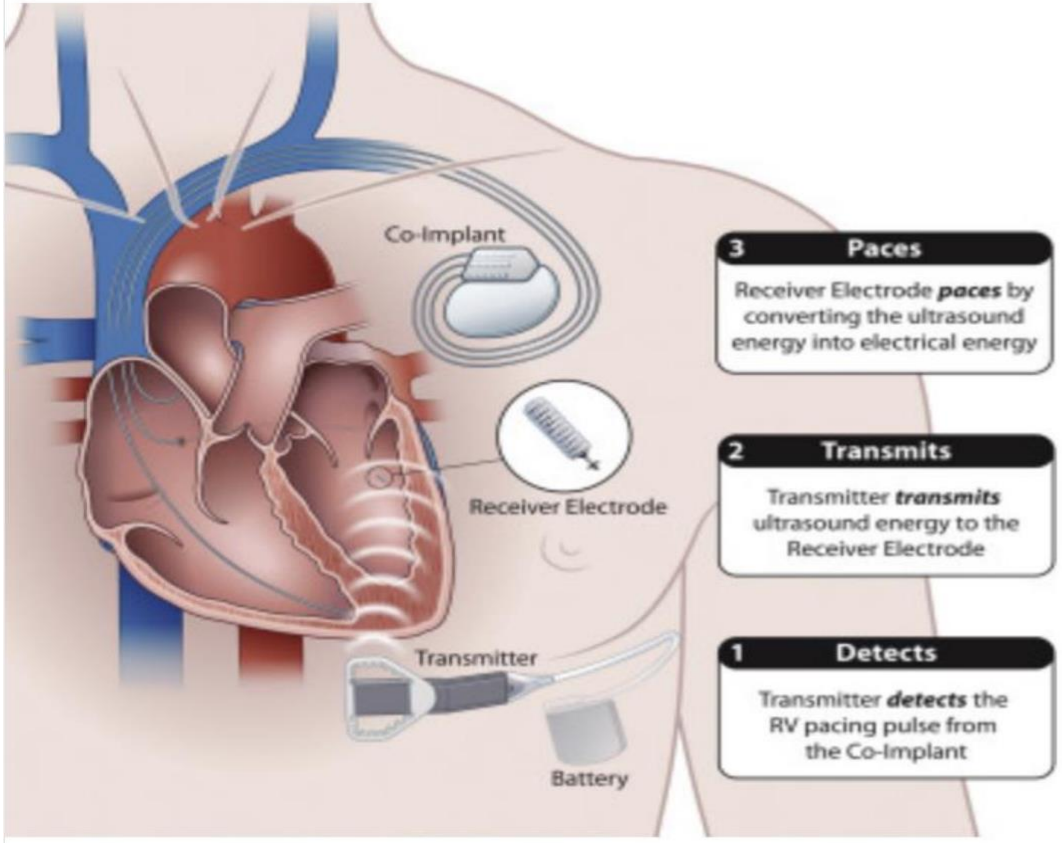
- Cardiomyopathy increase the risk of ventricular arrhythmias and sudden cardiac death
- Implantable cardiac defibrillator (ICD) for primary prevention (needs all 3):
  - ✓ GDMT for 3 months
  - ✓ LVEF of 35% or less
  - ✓ NYHA class II-III
- Secondary prevention:
  - ✓ Cardiac arrest secondary to VF or VT



Management – Invasive Interventions

Cardiac resynchronization therapy (CRT) indication:

- ✓ • GDMT for 3 months
- ✓ • LVEF of 35% or less
- ✓ • QRS duration of > 150 ms & left bundle branch block on ECG

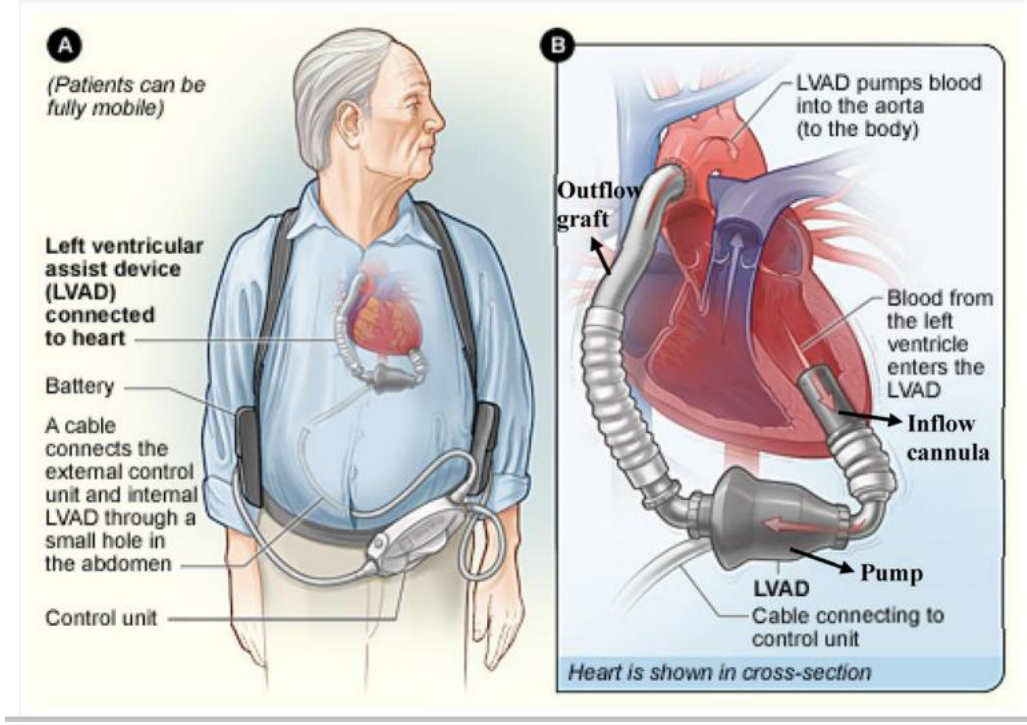




Management – Advanced HF

Cardiac transplant is the only cure for end-stage heart failure

- 1-year & 5-year survival rates are excellent
- Paucity in organ availability
- Bridging measures, including inotropic and mechanical circulatory support
- Refer patients who are not candidates for transplant to palliative care



University of Maryland School of Medicine Faculty Scientists and Clinicians Perform Historic First Successful Transplant of Porcine Heart into Adult Human with End-Stage Heart Disease

January 10, 2022 | Deborah Kotz



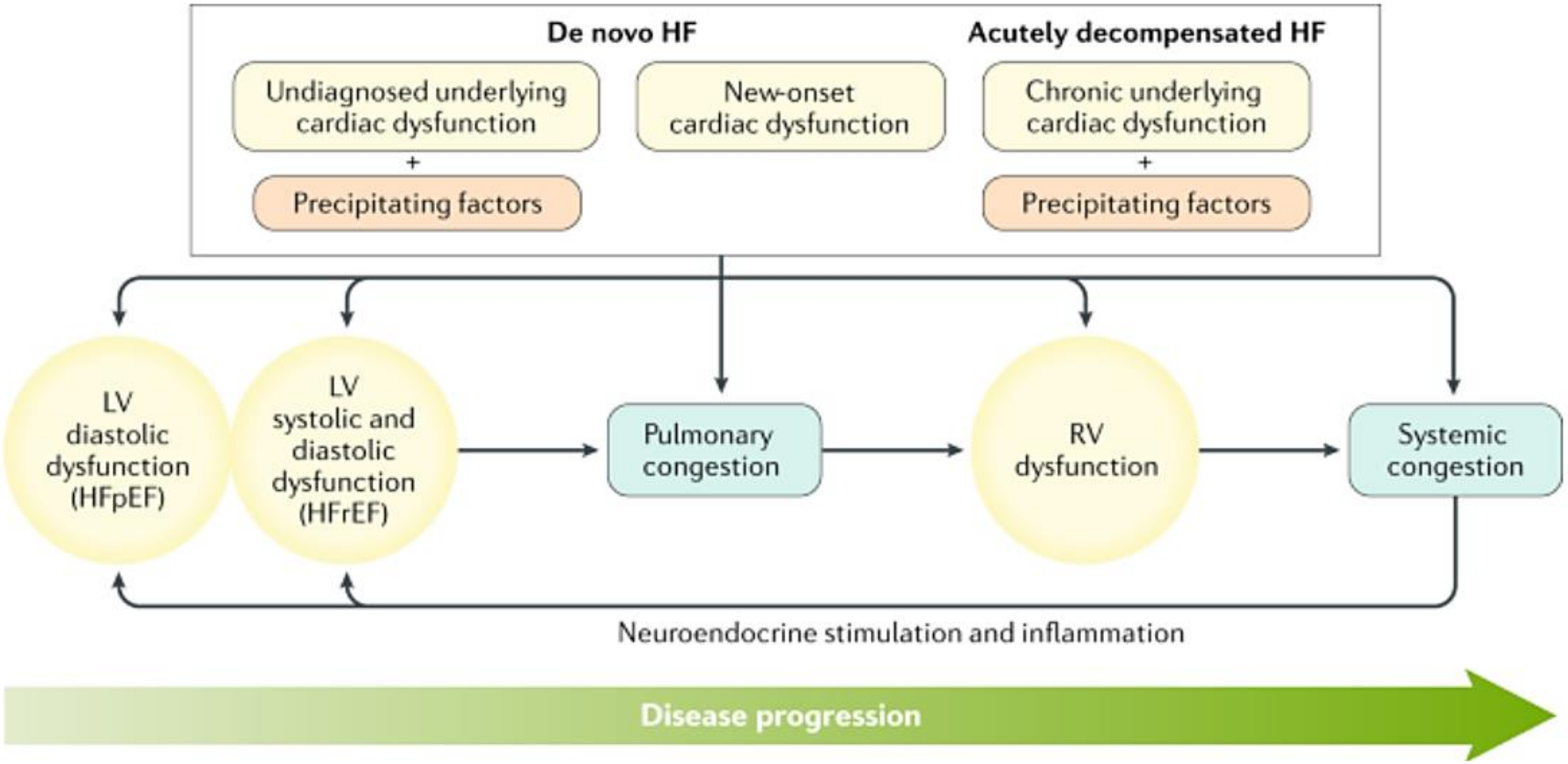
Subtypes & Variants – High Output HF

- HF secondary to conditions associated with a high-output state
- ↑ CO to meet the peripheral tissue oxygen demands
- Can be physiological (pregnancy)
- Can be pathological (severe anemia, arteriovenous fistula, Sepsis, beriberi & thyrotoxicosis)
- Treatment is directed at **correcting the underlying etiology**

Decreased peripheral vascular pressure (Afterload)	Increased metabolic demand (HR, contractility, afterload)	Mixed (peripheral vascular decrease + direct myocardial effect) (HR, afterload, contractility)	Other
• Morbid obesity • AV Fistula • Cirrhosis • Erythroderma • Carcinoid syndrome	• Myeloproliferative disorders • Hyperthyroidism	• Sepsis • Acromegaly • Mitochondrial diseases • Beriberi	• Anemia • Chronic pulmonary disease

Acute HF – Medical Emergency

- **Acute decompensated heart failure (ADHF):** acute heart failure due to decompensation of preexisting disease (most common)
- **De novo heart failure:** acute heart failure occurring for the first time in a patient without known cardiomyopathy



Acute HF – Medical Emergency

Triggers in ADHF:

- Medication non-compliance
- Ischemic heart disease
- Arrhythmias
- Serious infection/sepsis
- Drugs (NSAID, cocaine)
- Anemia
- Renal failure
- Volume overload

Acute HF – Medical Emergency

Triggers in De Novo:

Acute myocardial dysfunction

- Cardiac ischemia from ACS
- Myocarditis
- Drug-induced cardiomyopathy
- Peripartum cardiomyopathy
- Thyroid storm

Acquired valvular pathology

- Acute mitral regurgitation
- Bacterial endocarditis

Acute HF – Medical Emergency

Clinical Features:

Congestion

- Acute dyspnea and orthopnea
- Flash pulmonary edema
- Increased work of breathing
- Cough
- Coarse crackles/rales on auscultation

Hypoperfusion

- Weakness, fatigue, altered mental status
- Signs of poor peripheral perfusion

Acute HF – Medical Emergency

DDx:

- Acute coronary syndrome
- Pneumonia
- COPD, asthma
- Noncardiogenic pulmonary edema (ARDS)
- Pulmonary embolism
- Transfusion-related acute lung injury (TRALI)
- Transfusion associated circulatory overload (TACO)
- COVID-19

Acute HF – Medical Emergency

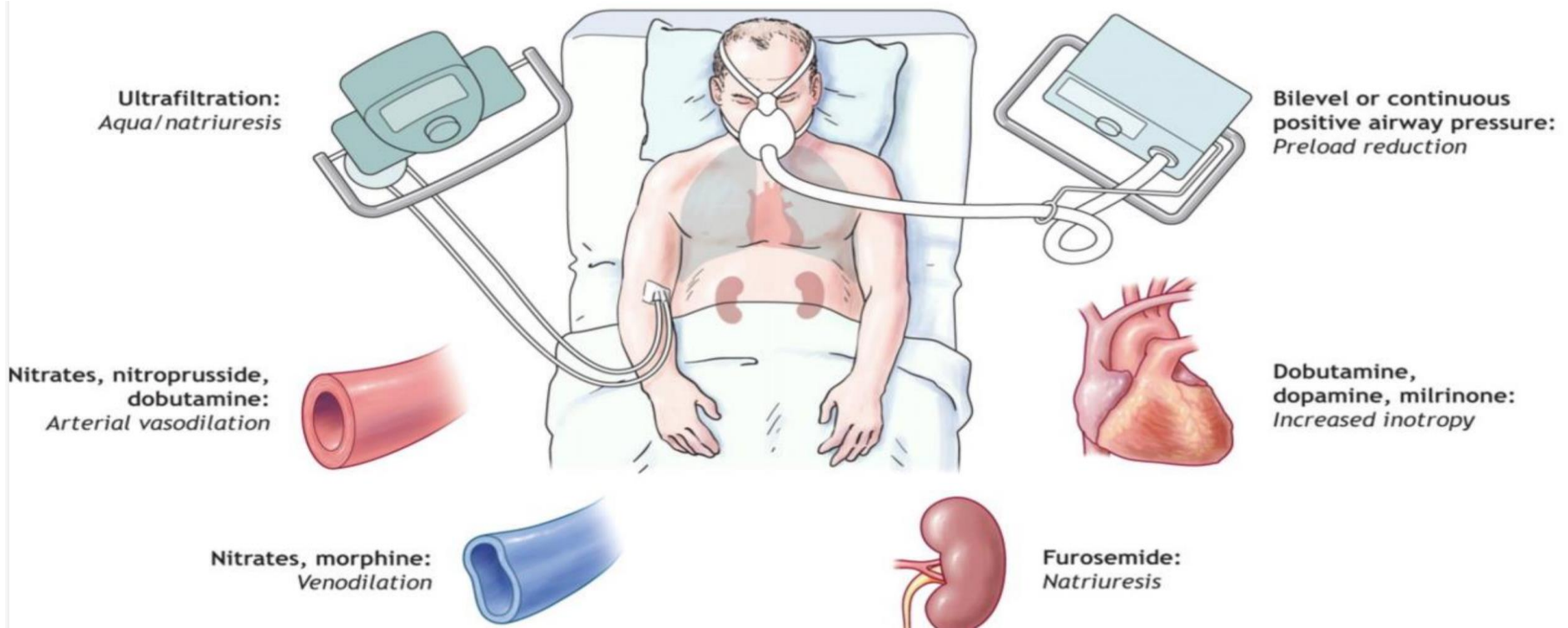
Diagnostics:

- BNP (or NT-proBNP)
- Troponin
- BMP and serum electrolytes
- CBC
- LFTs
- ECG
- CXR
- TTE
- COVID-19 & Flu swab
- TSH (in the right clinical setting)

Acute HF – Medical Emergency

Management:

- Respiratory support (Upright position, supplemental O2, NIPPV, Mechanical ventilation)
- Diuretic therapy (IV, monitor UOP and electrolytes)
- Vasodilator therapy (IV nitroglycerin, Sodium nitroprusside)
- Address exacerbating factor
- Once stable enough, optimize medical regimen with GDMT & evaluate for etiology)



References & further reading

- Harrison's Principles of Internal Medicine, 20th Edition
- AMBOSS
- American Heart Association (AHA) guidelines

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Thank You